AI-Based Sign Language Recognition System

1. Introduction

I chose this project because I wanted to learn how Artificial Intelligence can help improve communication for deaf and hard-of-hearing individuals. The project focuses on recognizing American Sign Language (ASL) using Machine Learning, Neural Networks, and Computer Vision techniques.

The system uses a live camera to detect hand signs, then translates the recognized English signs into Arabic text and Arabic speech. ASL datasets were used because they are more available and easier to work with than Arabic sign language datasets.

2. Dataset Description

The dataset contains hand landmarks extracted using MediaPipe instead of raw images. Each hand is represented using 21 landmarks with x, y, and z coordinates.

Total features:

21 \times 3 = 63

The dataset includes:

- ASL words
- ASL alphabet letters

Examples:

- hello
- thanks
- yes
- no
- A
- B
- C

The dataset was cleaned and checked for missing values before training. The data was then split into:

- 80% training data
- 20% testing data

3. Models Used

Three Machine Learning models were trained and compared:

- Logistic Regression
- Random Forest
- K-Nearest Neighbors (KNN)

The models were evaluated using:

- Accuracy
- Precision

- Recall
- F1-score

The best model was Logistic Regression with:

- Accuracy: 92.29%
- F1-score: 92.15%

It performed better because the landmark data is numerical and well-structured, making it suitable for traditional Machine Learning algorithms.

4. Neural Network

A Neural Network was built using Keras Sequential API. The model included:

- Dense layers
- ReLU activation
- Dropout layers
- Softmax output layer

The network was trained for more than 10 epochs. Training and validation accuracy curves were used to monitor performance and reduce overfitting.

Although the Neural Network learned complex patterns, Logistic Regression achieved better overall results on the test set.

5. Results & Comparison

After comparing all models, Logistic Regression achieved the highest performance and fastest prediction speed for real-time use.

The final system uses:

- OpenCV for camera input
- MediaPipe for hand detection
- Logistic Regression for prediction
- Arabic translation and Text-to-Speech output

The system can:

- Recognize ASL words and letters
- Translate predictions into Arabic
- Convert Arabic text into speech
- Build simple sentences from predictions

6. What I Learned

Through this project, I learned how to:

- Use Machine Learning and Neural Networks
- Process landmark data using MediaPipe
- Compare different AI models
- Build a real-time Computer Vision system
- Connect AI predictions with Arabic translation and speech output

I also learned that more complex models are not always better. In this project, Logistic Regression performed better than the Neural Network because the landmark features were already highly structured and organized.